

RABINA TWAYANA

MSc Student

Paris Lodron University Salzburg, Austria

E-mail: rabinatwayana123@gmail.com

[Linkedin](#) | [Github](#)

PROFESSIONAL SUMMARY

I am a certified geomatics engineer registered in Nepal Engineering Council, holding a four-year academic degree in Bachelor of Engineering in Geomatics from the Kathmandu University, Nepal. With four years of professional experience in the geospatial sector, I am currently pursuing Copernicus master's in digital Earth at Paris Lodron University Salzburg, Austria. I worked two years as a Research Associate at GeoInformatics Center, Asian Institute of Technology, Thailand. Prior to this, I worked two years as a WebGIS developer at Naxa Pvt. Ltd. and served as a Geospatial Analyst at Airlift Company both based in Kathmandu, Nepal. During my professional journey, I have contributed to diverse projects for esteemed organizations like University of Twente (UT) in the Netherlands, World Food Programme (WFP), World Bicycle Relief (WBF), Bhutan Construction and Transport Authority, Geodetic Survey Division of the Survey Department in Nepal, Clean-up Nepal, Changunarayan Municipality, and the Kathmandu Metropolitan City. This journey has equipped me with substantial expertise in WebGIS, geospatial programming, GIS, UAV technologies, remote sensing, and disaster management. I'm particularly driven by academic research and have published three articles in esteemed journals, one conference proceedings and one academic blog. Currently, my research interest lies in leveraging the potential of emerging geospatial technologies, including digital twin, 3D GIS, and geospatial programming, within the field of urban infrastructure and disaster management.

EDUCATION

Copernicus Master in Digital Earth

Oct 2024 - Present

Paris Lodron University Salzburg, Austria

Specialization: GeoData Science

Southern Brittany University, France

Bachelor's of Engineering in Geomatics

2016 - 2020

Kathmandu University, Dhulikhel, Nepal

Grade: 3.86 CGPA out of 4 (Distinction)

Thesis: Land Cover Change Analysis and its Future Prediction: A Case Study of Banepa and Dhulikhel Municipalities

PROFESSIONAL EXPERIENCE

Research Associate

Nov 2022 - Sept 2024

Geoinformatics Center, Asian Institute of Technology, Thailand

Project:

- Development of [RiskChanges](#) Platform, under Deliverable 4.1 (The web-based simulation and information service for multi-hazard impact chains), PARATUS Project, funded by European Union's Horizon Europe research and innovation programme

Major Responsibilities:

- Fixing bugs in existing components of application
- Brainstorming on integrating new features into application
- Designing and implementing new feature as per the project requirement
- Overseeing client-side, server-side, database handling and project server management
- Conducting testing, updating, and maintaining the project.
- Writing technical documentation related to application
- Contributing capacity building and training program

Full stack/WebGIS Developer**Dec 2020 - Oct 2022**

Naxa Pvt. Ltd., Kathmandu, Nepal

Projects:

- Re-design and development of Clean-up Nepal and mobile application [[link](#)]
- USAFIRI: An open-source, web-GIS based participatory mapping toolkit to aid evidence-based decision making [[link](#)]
- Development of Changunarayan Municipality Metric Addressing System
- Provincial Climate Change Management Information System (PCCMIS) [[link](#)]
- Design and development of a web-based application for Geodetic Division, Survey Department of Nepal
- Development of Open Data Portal for Changunarayan Municipality [[link](#)]
- Blue Infrastructure Mapping of Kathmandu Metropolitan City

Responsibilities:

- Participate in the overall application development lifecycle.
- Develop the server-side logic as well as the development and maintenance of the central database ensuring high responsiveness and performance to requests from the frontend.
- Engage in the development of high-quality frontend applications.
- Implement security measures in the developed software systems.
- Use proper version control tools and maintain standard release workflows for software system deployment.
- Carry out preliminary testing and quality assurance of the developed system before forwarding to QA team.
- Coordinate with business analysts, team members and client representatives in smooth development of the systems
- Engage in knowledge sharing and co-learning among team members in the company.

Geospatial Analyst**Feb 2023 – July 2023**

Airlift Company, Kathmandu, Nepal

Projects:

- Galli Maps Mobile Application Development
- Bhutan Road Corridor Design

Responsibilities:

- Processing UAV images data using Agisoft software.
- Revising the projects geospatial data using ArcGIS Desktop and QGIS software.
- Laise with clients as advised by project manager.

Frontend/WebGIS Developer Intern**Jun 2020 - Aug 2020**

Naxa Pvt. Ltd., Kathmandu, Nepal

Project:

- Headquarter Mapping (“An Open Spatial Data Initiative to Locate and Map 753 Local Governments and their Ward Offices in Nepal”)

Responsibilities:

- Develop the frontend (UI) using ReactJS library and interactive/dynamic map interface using Openlayers (JavaScript mapping library).

SKILLS

Geospatial Technologies: WebGIS, Remote Sensing, GIS, UAV, GNSS, GEE
Programming Language: Python
Web Development: HTML, CSS, JavaScript
Frontend Development: React, Redux
Backend Development: Django, Django REST Framework
Database Management: PostgreSQL
Geospatial Libraries: Openlayers, Leaflet, Cesium
Version Control: Git, Github
Containerization: Docker, Docker Hub
Geospatial Software: ArcGIS, QGIS, ENVI, GNSS solutions, Agisoft, Geoserver
Cloud Computing: Google Cloud Platform (GCP)
Operating System: Linux, Windows, Mac
Language: English, Nepali (Mother Tongue)

TRAINING AND PARTICIPATIONS

Sustainability Expo held on 23-24 September at Asian Institute of Technology, Thailand

Planet on the Road: Bangkok, 26 September 2023 at Bangkok, Thailand

Enhancing Inclusive Local Disaster Reduction Strategies Using Disability Inclusion Score Index, Online, 11 October 2023

WISESS Summer School in Risk and Resilience of 6 ECTS held on 10-14 July 2023 at University of Vienna, Austria

ASEAN Training on ICT for Climate Resilient Development, organized by APCICT, ESCAP held on 15-17 May 2023 Bangkok, Thailand

Dynamic Multi-Hazard Risk Assessment for Risk Reduction Planning, held on 10-15 Dec 2022 at Geoinformatics Center, Asian Institute of Technology, Thailand

Training on ArcGIS Desktop Software, held from 13-28 May 2018, organized by Geomatics Engineering Society, Kathmandu University, Dhulikhel, Nepal

VOLUNTEERING EXPERIENCE

Facilitator in Flood Risk Assessment Training for Bhutan's National Center for Hydrology and Meteorology, held from 21st Aug to 1st Sept 2023 at Geoinformatics Center, Asian Institute of Technology, Thailand

Editorial Board Member, Annual Geo-ITC Journal of Geomatics Engineering Society (ISSN 2091-198), Kathmandu University, Nepal

Organizing Committee Member, World GIS Day-2018, Kathmandu University, Nepal

HONOURS AND AWARDS

Erasmus Mundus Scholarship, 2024

Awarded full scholarship for joint master's program, Copernicus Master in Digital Earth at Paris Lodron University Salzburg, Austria and specialization in GeoData Science at Southern Brittany University, France

Nepal Institute of Chartered Surveyors (NICS), 2021

For securing second most position in Bachelor of Geomatics Engineering degree examination conducted by all Universities of Nepal

Kathmandu University

Waved 100% tuition fee in four semesters during undergraduate study for scoring highest GPA in respective semester

Land Management Training Centre (LMTC), Government of Nepal, 2016

Waved 35 % of tuition fee by LMTC for four years undergraduate course (BE in Geomatics) at Kathmandu University, Nepal.

Private and Boarding School's Organization Nepal (PABSON), District Executive Committee, Kavrepalanchowk, 2014

For securing distinction division with 86.38% marks in the SLC Exam

Nepal Student Union, District Executive Committee, Kavrepalanchowk, 2014

For securing highest position within school in SLC Exam

All Nepal National Free Students' Union (ANNFSU), Kavre District Committee, 2014

For securing highest position within school in SLC Exam

JOURNAL PUBLICATIONS

Twayana, R.; Bhandari, S.; Shrestha, R.; Sharma, K. (2021). *Analyzing Urban Growth Pattern and Driving Factors Using Remote Sensing and GIS: A Case Study of Banepa Municipality, Nepal*, Journal on Geoinformatics, Nepal, Vol. 20, No. 1 [[link](#)]

Shrestha, R.; Bhandari, S.; **Twayana, R.,** (2021), *Monitoring Land Cover/Land Use Change Using Remote Sensing and GIS Technique at Dhulikhel and Banepa Municipality, Nepal*, International Research Journal of Modernization in Engineering Technology and Science, Volume 03, Issue 05, Impact Factor- 5.354 [[link](#)]

Twayana, R.; Bhandari, S.; (2019), *Problems and its Possible Solutions for Conversion of 2D Cadastre to 3D Cadastre in Context of Nepal*, Geospace-Annual Geo ICT Magazine of GES, Vol. V, Page number 26 [[link](#)]

CONFERENCE PROCEEDING AND PRESENTATION

Presentation on *RiskChanges: A WebGIS Application for Multi-Hazard Quantitative Risk Assessment*, 3rd International Symposium on Disaster Resilience and Sustainable Development 2023, held on 7-8 Dec 2023, Asian Institute of Technology, Thailand

Bhandari, S.; Twayana, R.; Shrestha, R.; Sharma, K. (2021). *Future Land Use Land Cover Scenario Simulation Using Open-Source GIS for the City of Banepa and Dhulikhel Municipality, Nepal*, FOSS4G Asia 2021

ACADEMIC BLOG

An Open-Source Tool to Assist in Multi-Hazard Risk Assessment, published on 17 October 2023 in PreventionWeb [[link](#)]

PROFESSIONAL MEMBERSHIP

Women in Geospatial Mentorship Programme
Membership Type: Mentee

International Society of Photogrammetry and Remote Sensing (ISPRS)
Membership Type: Individual

International Student Nepal Engineering Council (NEC)
Membership Type: Geomatics 'A' Category
Membership Number: 439

Nepal Geomatics Engineering Society (GES) Member
Membership Type: General

REFEREES

Associate Prof. Dr. Reshma Shrestha
Head of Department
Department of Geomatics Engineering
Kathmandu University, Dhulikhel, Nepal
Email: reshma@ku.edu.np
Contact: +977 9841295224

Ph.D. Manzul Kumar Hazarika
Director
GeoInformatics Center (GIC)
Asian Institute Technology (AIT), Thailand
Email: manzul@ait.asia
Contact: +66 25246184

Er. Upendra Oli
Chief Technical Officer (CTO)
Naxa Pvt. Ltd., Kathmandu, Nepal
Email: upendra@naxa.com.np
Contact: +977 9841595614